HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Android framework customizations** for the Telephony and ConnectivityService layers, integrating with vendor-specific RIL.
- Optimized **LTE/5G network connectivity** performance, focusing on seamless handover and stability in mission-critical communication scenarios.
- Debugged complex system-level issues using **ADB and Wireshark**, resolving latency and connectivity drops to improve overall device performance.
- Developed and maintained **Java code for Android internals**, including JNI interfaces and framework modifications, ensuring robust functionality.
- Participated in **code reviews and technical estimations** within an Agile Scrum team, contributing to sprint planning and delivery timelines.
- Mentored junior engineers on **Android OS customization techniques** and best practices for AOSP development.
- Containerized Android build and testing environments using **Docker**, enabling consistent development workflows across distributed teams.

SKILLS

Android – AOSP, Android Framework, Telephony Framework, ConnectivityService, Hardware Abstraction Layers, RIL, ADB, dumpsys, tombstone

Languages – Java, C++, JNI

Wireless Protocols – LTE, 5G, NAS, AS, RRC

Build Systems – Soong, Ninja, Make, Gradle

Tools & Methodologies – Git, Gerrit, Jira, Wireshark, qxdm, Performance Tuning, Debugging, Agile, Scrum

Cloud & Containerization – Docker, Linux

PROJECTS

AOSP Customization for LMR Devices

- Modified and extended **Android framework layers**, including the Telephony stack, to support specific hardware requirements for land mobile radio devices.
- Integrated vendor-specific **Radio Interface Layer (RIL)** and modem firmware, ensuring stable and reliable network communication.
- Performed system-level debugging and **performance tuning** on test devices, resolving connectivity issues and optimizing CPU/memory usage.

LTE Connectivity Optimization Framework

- Developed a framework for fine-tuning **LTE performance**, focusing on handover stability and seamless connectivity in challenging network conditions.
- Analyzed network traffic using **Wireshark** and QXDM to identify bottlenecks and implement optimizations for critical communications.
- Collaborated with hardware teams to validate RIL integrations and modem firmware updates, ensuring adherence to performance targets.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)